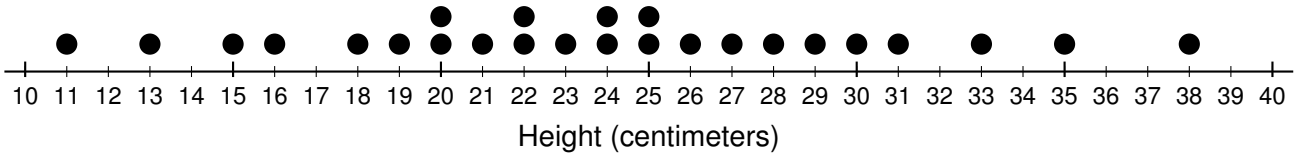
Unit 1 Test: Displaying One Variable

Name _____

Show your work.

1. **I make a histogram.**

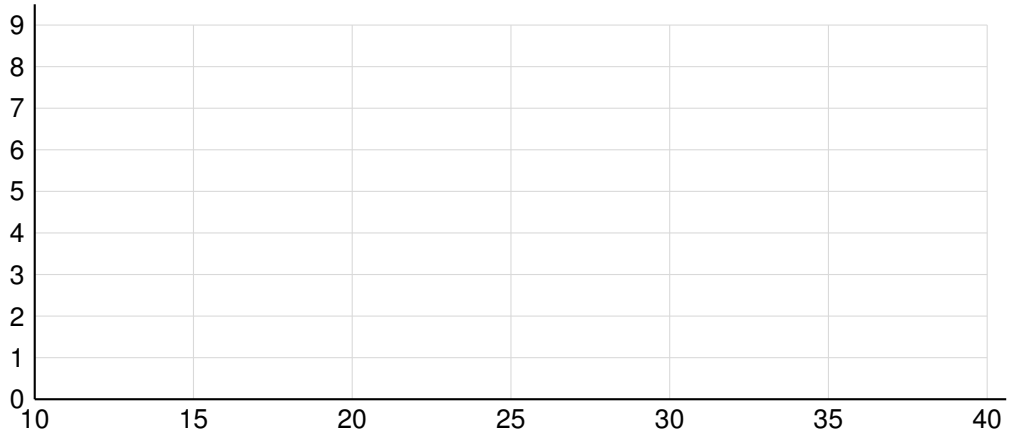
   A class grew 25 bean plants. The dot plot shows the height of each plant.



   (a) Make a frequency table. Count how many plants are in each group. Remember the histogram rule.

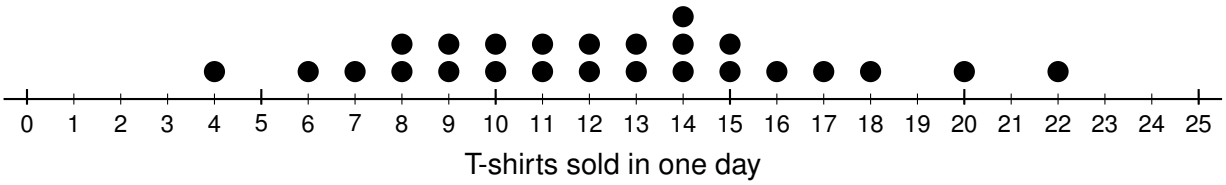
Height	Frequency
10 to 15	
15 to 20	
20 to 25	
25 to 30	
30 to 35	
35 to 40	

   (b) Make a histogram. Label both axes.



2. **I find the five-number summary, and I make a box plot.**

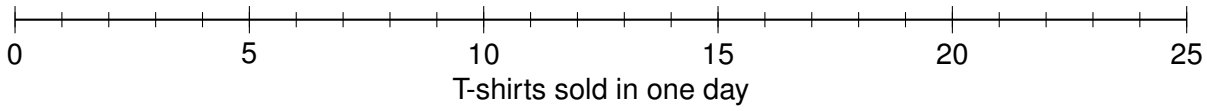
A store kept track of T-shirt sales for 25 days. The dot plot shows how many T-shirts it sold each day.



(a) Find the five-number summary.

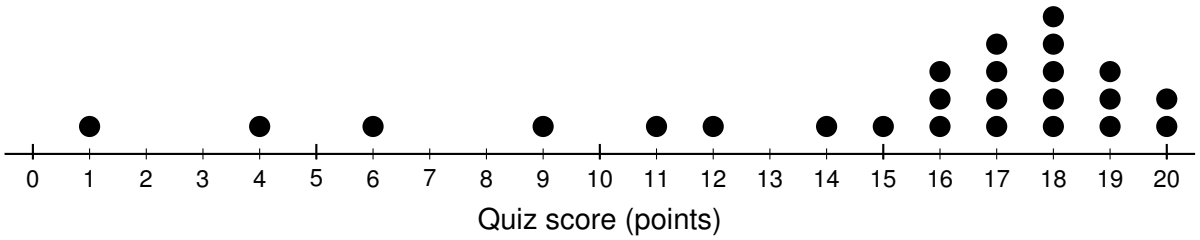
Min _____ Q1 _____ Median _____ Q3 _____ Max _____

(b) Make a box plot.



3. **I find the mean and the median, I tell the shape, and I choose the best measure of center.**

25 students took a quiz. The quiz had 20 points. The dot plot shows each student's score.



(a) Find Σx and the mean, $\bar{x}$. (Σ means add up. n is the number of values.)

(b) Find the median.

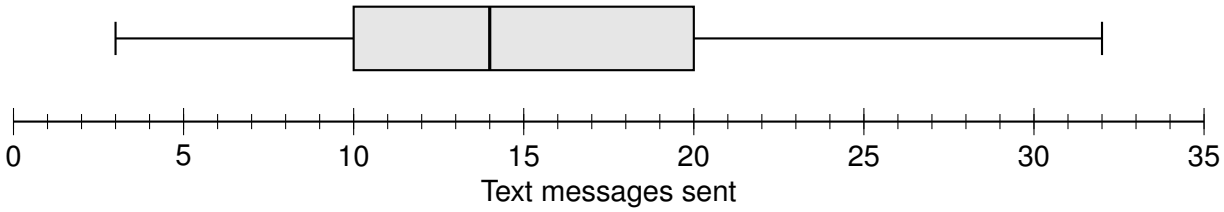
(c) Circle the shape of the data.

skewed left symmetric skewed right

(d) Which describes a typical score better, the mean or the median? Explain.

4. **I use the 1.5 $\times$ IQR rule to find outliers.**

The box plot shows how many text messages 25 students sent yesterday.



One more student sent 60 text messages. Now there are 26 students.
Adding 60 does **not** change Q1, the median, or Q3.

(a) Find the IQR. Then find the lower fence and the upper fence.

(b) Is 60 an outlier? Explain.

(c) How would the box plot change? What would stay the same? You do not have to draw a new box plot.